

doubts

✓
✓
exercise int
32 bits.

$\{-1, 0\}$ xor = -1

$-1 \rightarrow \text{flip} \rightarrow 110 \rightarrow 110 + 001 = 111$
 $1 \rightarrow 001$

xor

	1	1	1
1	0	0	0
MSB	1	1	1

$\rightarrow -ve no.$

$(111) \rightarrow 000 + 1 = 001$
xor = $-(001)$

$0 \dots n-1$ } $\rightarrow$ linear search
 $K \rightarrow \underline{\underline{i \rightarrow \text{index}}}$

0 ... n-1

$$K \rightarrow \oplus$$


searching?

how?

K=6 →

search

find

4 → ans

$$\begin{bmatrix} 1 & 5 & 2 & 4 & 6 & 3 & 9 \\ 0 & 1 & 2 & 3 & 4 & & \end{bmatrix}$$

searching

$$\textcircled{I} \rightarrow [I_1, I_2, I_3, I_4]$$

Time complexity $\rightarrow$ WC

A hand-drawn diagram of a linked list. It shows three nodes: a top node labeled n , a middle node labeled $n-1$, and a bottom node labeled $n-2$. Arrows indicate the sequence: from n to $n-1$, and from $n-1$ to $n-2$. The nodes are represented by rectangles with labels inside, and the arrows are drawn outside the rectangles.

$$\frac{n}{K} \approx O(n)$$



if(flag == false)
print(-1);

$K=5$ $n=6$
 $arr = [1, 2, 4, 9, 6, 5]$
flag = false;
for($i=0$; $i < n$; $i++$) { $\approx O(n)$
 if($arr[i] == K$) {
 print(i);
 flag = true;
 break;
 }
}

$O(n^3)$

$n=1000$

1 - 2 5 - 3 2 4 → [1 5 2 3 9]

added

Kadane's
algo

max subarray sum
?

gen all subarrays

curr → sum

final ← (ans)

st
 $i=0 \dots n-1$
 $j=i \dots n-1$
 $k=i \dots j$

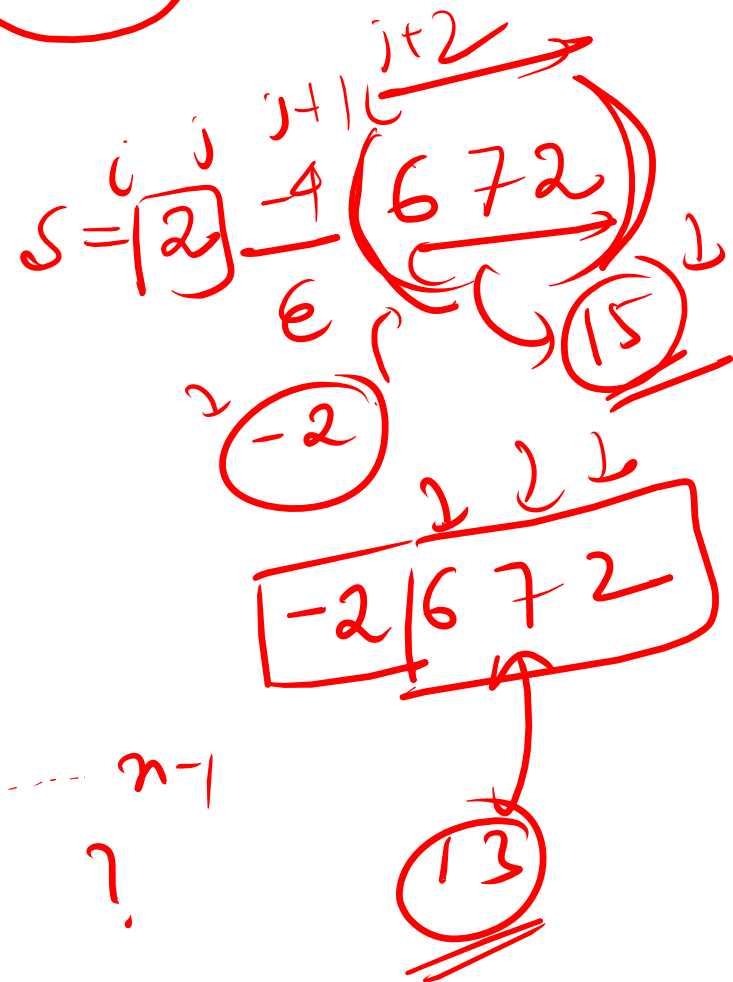
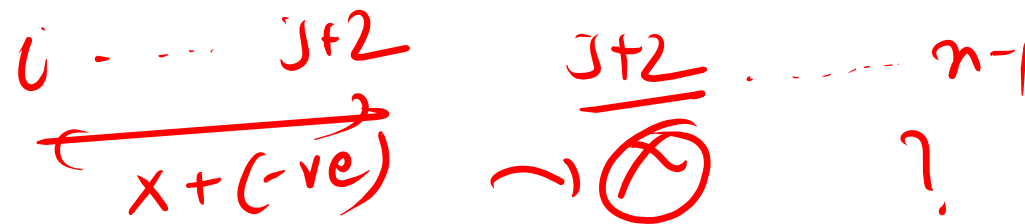
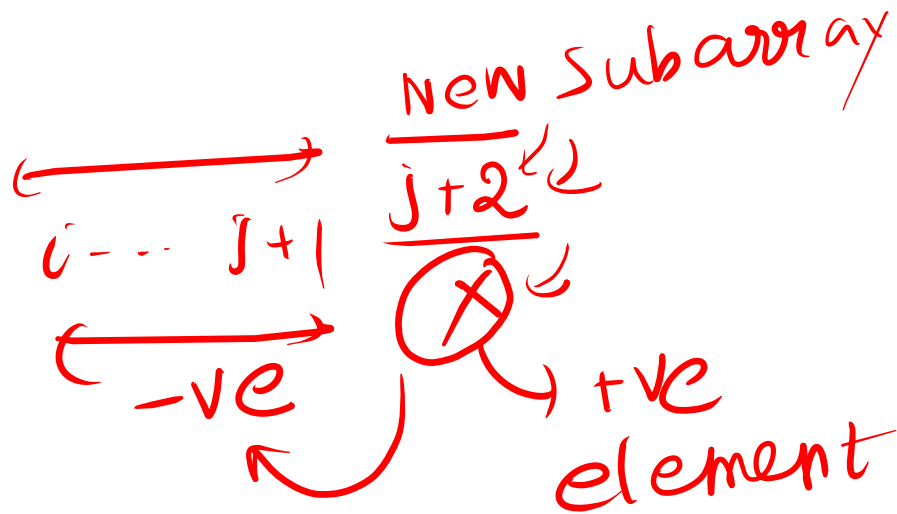
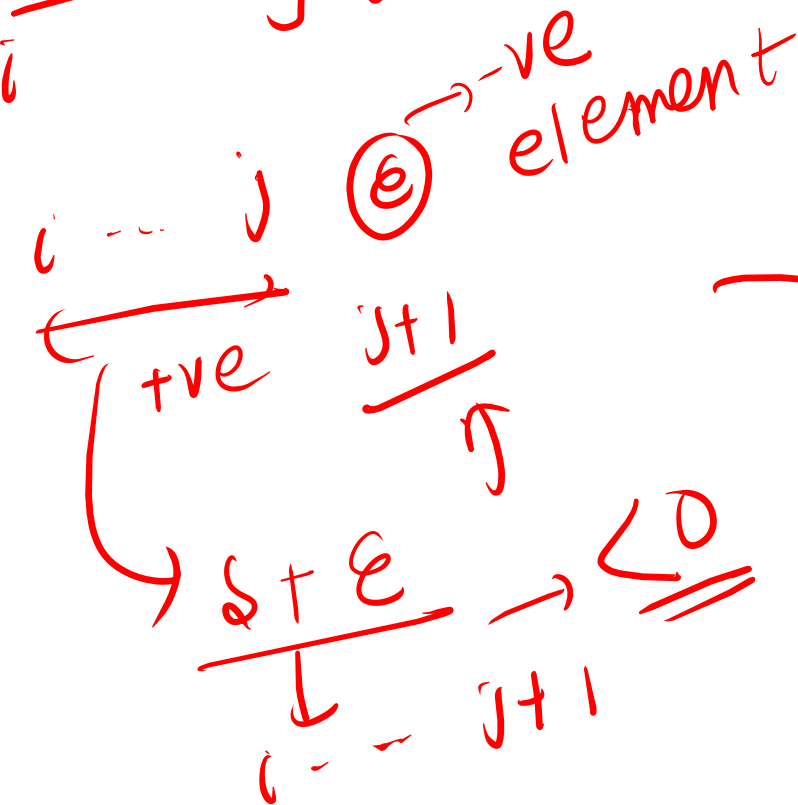
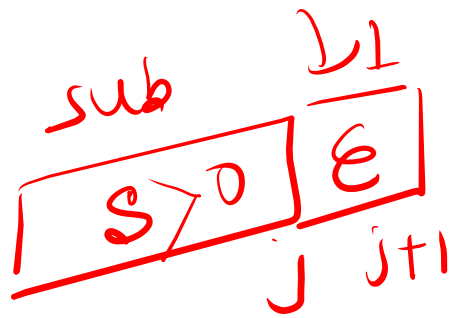
sum += arr[k]

ans = max(ans, sum);

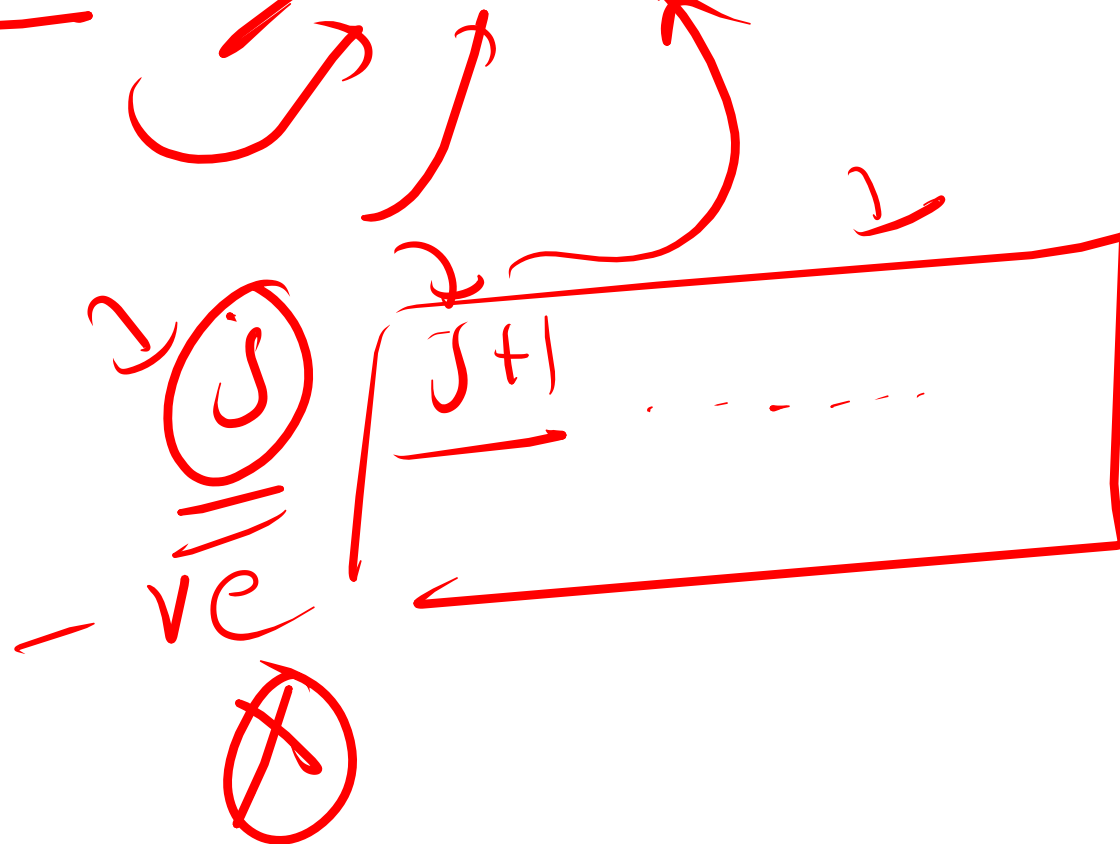
ans = final

1 - 2 5 - 3 2 4

15 $\rightarrow$ max / 13
 (+ve) + (-ve)



curr $\rightarrow$ ~~-ve~~ 0



$curr = \emptyset$ 1
 $ans =$
~~(INT_MIN)~~
~~1~~ ~~8~~
 $1 - 2 \begin{pmatrix} 5 & -3 & 2 & 4 \end{pmatrix} \oplus$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 4 & 5 \end{pmatrix} \rightarrow$ once $O(n) \rightarrow TC$
8

$\begin{matrix} & & j & j+1 & j+2 \\ & & \downarrow & & \downarrow \\ \begin{pmatrix} i & & & & \end{pmatrix} & \begin{pmatrix} s & \end{pmatrix} & \begin{pmatrix} e & \end{pmatrix} & \begin{pmatrix} x & \end{pmatrix} & \begin{pmatrix} x & \end{pmatrix} \end{pmatrix} \rightarrow x > 0$
 $st e < 0$
 $-ve$

$curr = 1 - 2$
 $\begin{pmatrix} & & & & & & \end{pmatrix} = \cancel{1} \cancel{\emptyset} \begin{pmatrix} 5 & -3 \end{pmatrix}$
 $= \underline{2} + 2$
 $= 4 + 4 = \underline{8}$

$s = \underline{0} + \cancel{8}$
ans